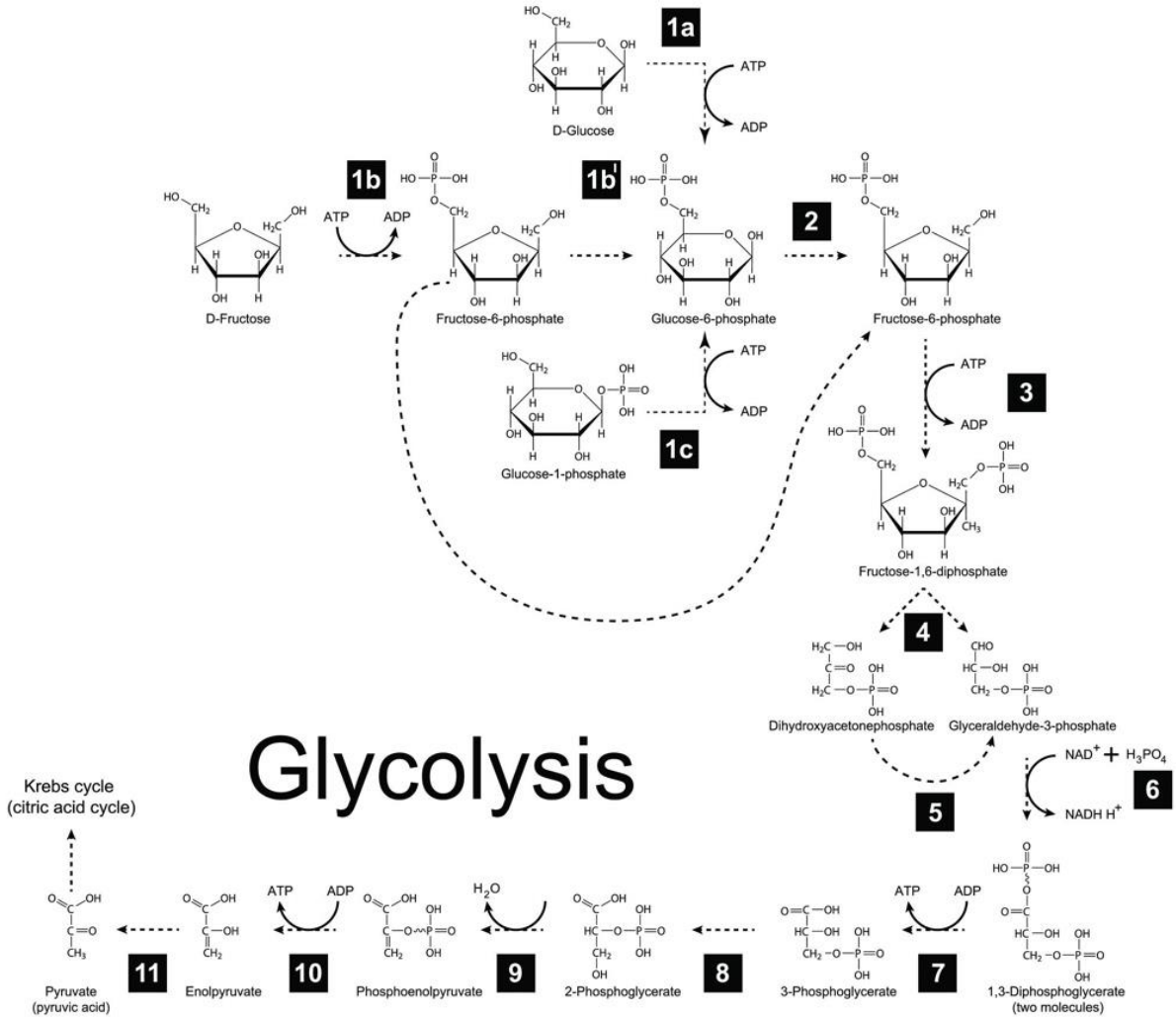
Biochemistry Cume Question

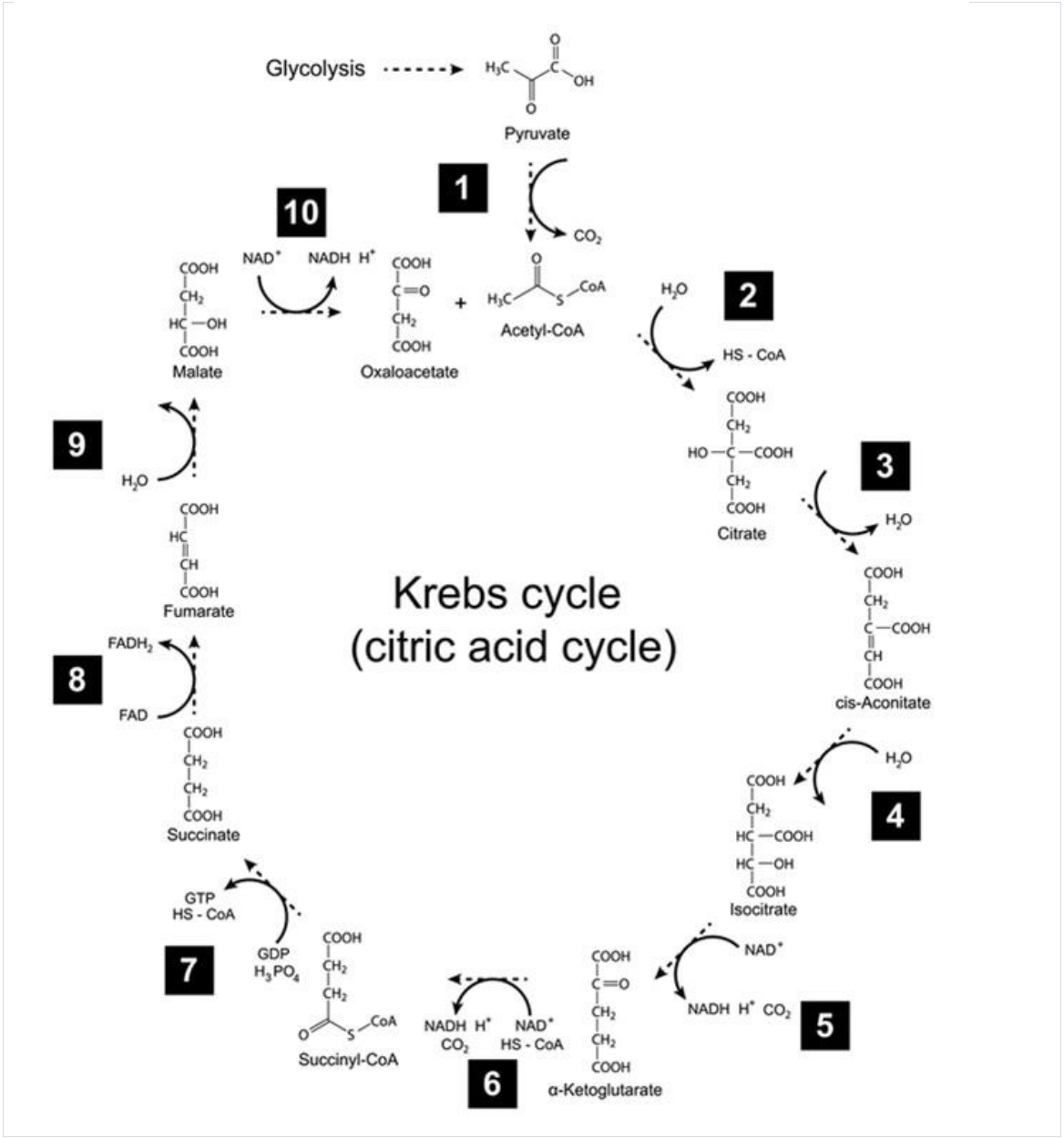
Understanding metabolic pathways is crucial in fields like medicine and biotechnology, as it helps explain diseases caused by metabolic defects and guides the development of therapeutic interventions.

Question:
Trace the fate of carbon atom number 2 (C2 labeled with ^{14}C) in a pyruvate molecule as it undergoes the citric acid cycle (TCA cycle, Krebs cycle) and is eventually converted to aspartate (Asp) through the process of transamination. Provide a step-by-step account of the relevant reactions and intermediates involved in the TCA cycle and the transamination process. Label the location of ^{14}C in the intermediates.

Instructions:

- Clearly indicate how the ^{14}C-labeled carbon atom progresses through each step of the citric acid cycle.
- Identify the point at which the carbon atom is incorporated into aspartate through the transamination reaction.


Glycolysis
1a
D-Glucose
ATP
ADP
1b
D-Fructose
1b'
Fructose-6-phosphate
Glucose-6-phosphate
2
Fructose-6-phosphate
1c
Glucose-1-phosphate
3
Fructose-1,6-diphosphate
4
Dihydroxyacetonephosphate
Glyceraldehyde-3-phosphate
5
6
NAD+ + H3PO4
NADH H+
1,3-Diphosphoglycerate
(two molecules)
7
3-Phosphoglycerate
8
2-Phosphoglycerate
9
H2O
Phosphoenolpyruvate
10
Enolpyruvate
11
Pyruvate
(pyruvic acid)
Krebs cycle
(citric acid cycle)


Glycolysis
Pyruvate
1
CO_2
Acetyl-CoA
2
H_2O
HS - CoA
Citrate
3
H_2O
cis-Aconitate
4
H_2O
Isocitrate
5
NAD^+
NADH H^+ CO_2
α-Ketoglutarate
6
NADH H^+
CO_2
NAD^+
HS - CoA
Succinyl-CoA
7
GDP
H_3PO_4
GTP
HS - CoA
Succinate
8
$FADH_2$
FAD
Fumarate
9
H_2O
Malate
10
NAD^+
NADH H^+
Oxaloacetate
Krebs cycle
(citric acid cycle)